

Ege Demirci

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EDUCATION

- **University of California, Santa Barbara** Santa Barbara, USA
Doctor of Philosophy - Computer Science September 2024 - June 2028 (Expected)
Advisor: Ambuj K. Singh, Academic Excellence Fellowship
- **Sabanci University** Istanbul, Turkey
Bachelor of Science - Computer Science and Engineering; GPA: 4.0 / 4.0 September 2019 - June 2024
Graduated as 1st ranked student in the Faculty of Engineering (1/609), 100% Scholarship

PUBLICATIONS

- [1] **Demirci, E.**, Bullo, F., Swami, A., & Singh, A. (2026). *FlowSymm: Physics Aware, Symmetry Preserving Graph Attention for Network Flow Completion.* (ICLR 2026)
- [2] **Demirci, E.**, & Singh, A. (2026). *Concorde: Geometry-Aware Link Prediction via Decoupled Energy Minimization.* (NeurIPS 2026)
- [3] **Demirci, E.**, Kerur, R., & Singh, A. (2025). *Are LLMs Truly Graph-Savvy? A Comprehensive Evaluation of Graph Generation.* (ACL 2025 - SRW)
- [4] **Demirci, E.**, & Singh, A. (2026). *The Symmetry Stress-Test: Violation-Aware Benchmarking for Calibrated Equivariance.* (Under review for ICLR 2027)
- [5] **Demirci, E.**, Sardina, C., & Singh, A. (2026). *Circles or Orderings? A Symmetry Test for haCyclic Concepts in Language Models.* (Under review for ICLR 2027)
- [6] **Demirci, E.**, Tuzun, E., Un, A. F., Sonmez, T. G., & Varol, O. (2026). *From Occasional to Steady: Causal Drivers of Sustained Gym Attendance in a Large-Scale Fitness Study.* (EPJ Data Science; presented at IC2S2 2025)
- [7] Najafi, A., Mugurtay, N., Zouzou, Y., **Demirci, E.**, Demirkiran, S., Karadeniz, H.A., & Varol, O. (2024). *First public dataset to study 2023 Turkish general election.* *Scientific Reports*, 14(1), 8794.

INDUSTRY EXPERIENCE

- **Amazon** Seattle, USA
Applied Scientist Intern June 2026 - September 2026
 - Investigated systematic “false zeros” in a **large-scale production ML incrementality model** (regularized regression over a multi-channel marketing portfolio).
 - Built a **self-contained experimentation model** replicating the production pipeline (cutting per-experiment runtime from **hours to minutes**); used it to engineer **propensity features** validated against **randomized controlled trials**, reducing model-wide bias **~13%**.
 - Resolved a live cross-org escalation **four independent ways** (per-segment retraining, causal and raw-behavior analysis, attribution-window sweeps, and upstream exposure-quality modeling), showing a channel’s low value reflected *which customers it reached*, with roughly **80%** of its edge attributable to audience mix rather than causal lift.
 - Co-authored a **technical deep-dive** reviewed by **senior leadership**, informing model-design and productionization decisions.
- **AppFolio** Santa Barbara, USA
Software Engineer Intern June 2025 - September 2025
 - Led development of a new LLM-powered **Smart Search** feature from concept to launch; authored **3** technical design documents on AI integration while developing core features with **Ruby** and **React**.
 - Resolved user-reported UX issues, increasing the team’s average monthly UX bug fixes by **300%** through streamlined debugging processes.
 - Shipped multiple full-stack features for the Investment Management application, ranging from complex dashboard redesigns to backend performance optimizations.
- **Mars Athletic** Istanbul, Turkey
Research Scientist (Part-Time) September 2023 – August 2024
 - Led a large-scale study on exercise habit formation using machine learning and causal inference, analyzing data from **200,000+** members of Mars Athletic (Turkey’s largest gym chain). [6]
 - Coordinated project design and team efforts, introducing a novel survival metric to quantify habit durability.
 - Evaluated the impact of interventions (group classes, personal training, social interactions) on long-term exercise adherence, providing actionable insights for management and public health.
- **Commencis** Istanbul, Turkey
Software Engineer (Part-Time) July 2023 – August 2024
 - Core member of the 4-person squad responsible for the **Campaigns domain** within the Turkcell Superapp (35M+ users); delivered multiple features **end-to-end**, from architectural design to UI implementation in **Swift**.
 - Co-developed the redesign of the campaign discovery flow, building custom UI components and integrating enhanced conversion tracking to drive user engagement.
 - Optimized module performance by refactoring legacy code and resolving critical concurrency bugs, resulting in improved stability and reduced crash rates.

RESEARCH EXPERIENCE

- **DYNAMO Lab, UC Santa Barbara** Santa Barbara, USA
September 2024 - Present
Researcher, Advisor: Dr. Ambuj Singh
 - Developed **FlowSymm**, a graph attention architecture that integrates symmetry-preserving group actions with bilevel optimization for network flow completion, achieving up to **10% RMSE reduction** over prior state-of-the-art. [1]
 - Developed **Concorde**, a mixed-curvature link prediction framework that unifies hyperbolic and spherical geometry with local diffusion, nearly **doubling early precision** on sparse graphs compared to leading GNN baselines. [2]
 - Built a **violation-aware benchmark** for equivariant models, showing calibrated symmetry holds accuracy under structured symmetry-breaking where strictly equivariant models collapse, at matched compute. [4]
 - Showed cyclic concepts in **large language models** are represented as measurable $SO(2)$ **rotations**, recovering the group action across **8 models (410M–12B)** and framing the Linear Representation Hypothesis as its flat limit. [5]
 - **Research Focus:** Graph Machine Learning, Equivariant & Geometric Deep Learning, LLM Interpretability
- **Viral Lab - VRL Lab, Sabanci University** Istanbul, Turkey
January 2022 - August 2024
Researcher, Advisor: Dr. Onur Varol
 - Analyzed misinformation and propaganda networks on X and YouTube during the 2023 Turkish elections. [7]
 - Built a scraper to collect X profiles of **1,500+** politicians active since 2010, creating the first public dataset of Turkish politicians with trending topics, media contacts, and follower dynamics.
 - Collected 1M+ YouTube comments, performed sentiment analysis, and studied causal links between media coverage and political polarization.
 - **Research Focus:** Applied Machine Learning, Computational Social Science, Network Science, Social Networks
- **Computational Intelligence Group, Vrije University** Amsterdam, Netherlands
June 2022 – September 2022
Research Intern, Advisor: Dr. Anil Yaman
 - Developed reinforcement learning agents for MuJoCo and Classic Control environments in GymLibrary.
 - Implemented neural network solutions for CartPole, Ant, Hopper, and Acrobot, and evaluated performance using standard RL benchmarks and metrics.
 - **Research Focus:** Reinforcement Learning, Artificial Neural Networks

TEACHING EXPERIENCE

- **University of California, Santa Barbara** Santa Barbara, USA
September 2025 - June 2026
Teaching Assistant
 - Lead weekly discussion sections and office hours, teaching Python and foundational data science concepts (CMPSC 5A) to undergraduates.
- **Sabanci University** Istanbul, Turkey
September 2021 - June 2024
Learning Assistant
 - Instructed recitations for **Network Science (CS414)**, **Intro to Data Science**, and **Python**, mentoring 200+ students over 3 years.

SKILLS SUMMARY

- **Languages:** Python, C++, C, SQL, JavaScript, Ruby, Swift
- **ML & Data Stack:** PyTorch, Scikit-learn, NetworkX, Pandas, Selenium, DGL, DoWhy
- **Software Engineering:** React, Ruby on Rails, LLM Integration, Node.js, Swift, SQL, PostgreSQL, Git
- **Product & Process:** Technical Writing, Experimental Design, A/B Testing, Agile Development, Cross-Functional Collaboration